

Notes: Belo, Lin, and Bazdresch (JPE, 2014)

Labor Hiring, Investment, and Stock Return Predictability in the Cross Section

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1 One-sentence summary

This note summarizes the paper, which uses CRSP/Compustat firm-level employment, investment, accounting, and stock-return data; hiring and investment portfolio sorts; Fama–MacBeth and pooled return-predictability regressions; CAPM and Fama–French asset-pricing tests; and a calibrated investment-based asset-pricing model with stochastic labor and capital adjustment costs to show that high-hiring firms earn lower subsequent average returns because expanding firms hedge aggregate adjustment-cost shocks.

2 Big picture

2.1 Research question

The paper asks whether labor-market frictions matter for asset prices in the cross section of firms.

- Traditional investment-based asset-pricing models emphasize physical capital adjustment costs: firms that invest a lot tend to earn lower future returns because investment reveals low required returns or favorable growth opportunities.
- Belo, Lin, and Bazdresch extend this logic to labor. Hiring workers is treated as an investment-like decision because adjusting the workforce is costly.
- The central empirical question is whether current firm-level hiring predicts future stock returns after accounting for standard investment effects.

2.2 Main empirical fact

Firms with high current hiring rates earn lower subsequent stock returns.

- In portfolio sorts, low-hiring firms outperform high-hiring firms by about **5.6 percent per year** in value-weighted returns and about **10.4 percent per year** in equal-weighted returns.
- In firm-level regressions, a 10 percentage point increase in the hiring rate predicts roughly a **1.5 percentage point decrease** in annual future returns.
- The hiring effect survives controls for physical investment, so labor hiring is not merely a relabeling of the investment anomaly.

2.3 Main theoretical answer

High-hiring firms are expanding firms. Because adjustment costs prevent immediate adjustment, these firms expect to keep hiring and investing. If an aggregate shock lowers adjustment costs, expanding firms benefit most: their future adjustment becomes cheaper, their continuation values rise more, and their returns covary positively with the adjustment-cost shock. Since the adjustment-cost shock carries a negative price of risk, high-hiring firms are hedges and therefore earn lower average returns.

3 Incremental contribution of the paper

3.1 Contribution relative to investment-based asset pricing

The paper adds labor adjustment to the investment-based asset-pricing framework. Earlier production-based models mainly use capital investment to explain return spreads. This paper

shows that labor hiring has its own cross-sectional pricing content.

3.2 Contribution relative to empirical anomalies

The paper documents a new return predictor: the firm-level hiring rate. It shows that the hiring return spread is distinct from the investment return spread, although the two are positively related in the data.

3.3 Contribution relative to labor economics and macro-finance

The paper connects labor frictions to equity risk premia. Labor adjustment costs, usually studied in labor demand and business-cycle models, are shown to affect firms' exposure to macroeconomic shocks and therefore their expected returns.

3.4 Contribution relative to factor models

The paper explains why the unconditional CAPM fails for hiring portfolios. The market factor mainly reflects aggregate productivity risk, while the hiring spread is driven by exposure to adjustment-cost risk. A multifactor model such as Fama–French performs better because it partially proxies for this additional risk.

4 Data and key variables

4.1 Sample

The empirical sample combines CRSP monthly stock returns with Compustat annual accounting data.

- Sample period: July 1965 to June 2010.
- Firms: common stocks listed on NYSE, AMEX, and NASDAQ.
- Exclusions: financial firms, regulated utilities, missing or invalid capital/employment/investment observations, and firms without required accounting information.
- Final sample: 75,381 firm-year observations.

4.2 Hiring rate

The hiring rate is defined symmetrically around zero:

$$HN_t = \frac{H_t}{0.5(N_{t-1} + N_t)}, \quad H_t = N_t - N_{t-1},$$

where N_t is the number of employees. This measure is bounded between -200% and 200% and treats hiring and firing symmetrically.

4.3 Investment rate

The investment rate is defined as

$$IK_t = \frac{I_t}{0.5(K_{t-1} + K_t)},$$

where K_t is net property, plant, and equipment, and I_t is capital expenditures net of sales of property, plant, and equipment.

4.4 Other firm characteristics

The paper also tracks size, physical-capital-to-market-equity ratio, profitability, and productivity.

- **Size:** market capitalization.
- **KM:** physical capital divided by market equity, related to book-to-market.
- **ROA:** net income divided by book assets.
- **TFP:** firm-level total factor productivity from Tuzel and Imrohoroglu.

5 Empirical design

5.1 One-way hiring portfolios

At the end of June of each year, firms are sorted into 10 portfolios based on the previous fiscal year’s hiring rate. Returns are measured from July of year t to June of year $t + 1$.

- Portfolio breakpoints are based on the distribution of hiring rates among all but micro-cap firms.
- The paper reports equal-weighted, value-weighted, and equal-weighted non-micro-cap returns.
- The low-minus-high hiring portfolio is the main hiring return spread.

5.2 Two-way hiring and investment portfolios

The paper also forms portfolios jointly sorted on hiring and investment.

- Firms are first sorted into three investment portfolios.
- Within each investment portfolio, firms are sorted into three hiring portfolios.
- This design tests whether hiring predicts returns within investment bins, and whether investment predicts returns within hiring bins.

5.3 Firm-level return-predictability regressions

The paper estimates regressions of future stock returns on lagged hiring and investment.

$$r_{it}^s = a + bHN_{i,t-1} + cIK_{i,t-1} + dMicro_i + e(Micro_i \times HN_{i,t-1}) + f(Micro_i \times IK_{i,t-1}) + \varepsilon_{it}.$$

The authors estimate both Fama–MacBeth cross-sectional regressions and pooled OLS regressions with firm and year fixed effects.

5.4 Asset-pricing tests

The paper estimates monthly time-series regressions of portfolio excess returns on:

- the market factor for CAPM tests;
- the market, SMB, and HML factors for Fama–French three-factor tests.

The main diagnostic is whether these models explain the low-minus-high hiring return spread.

6 Core empirical results

6.1 Hiring predicts future stock returns

The hiring spread is large and statistically significant.

- Equal-weighted low-minus-high hiring spread: about 10.4 percent per year.
- Equal-weighted excluding micro caps: about 6.9 percent per year.
- Value-weighted: about 5.6 percent per year.

The Sharpe ratios also decline strongly with hiring. Low-hiring portfolios have much higher Sharpe ratios than high-hiring portfolios.

6.2 High-hiring firms look like expanding firms

High-hiring firms tend to have:

- higher investment rates;
- higher productivity;
- higher profitability;
- lower capital-to-market-equity ratios.

This confirms that high hiring is economically meaningful: it captures expansion after favorable firm-level productivity shocks.

6.3 Hiring contains information beyond investment

The two-way portfolios and the firm-level regressions show that hiring and investment both predict returns. Within investment bins, low-hiring firms earn higher returns than high-hiring firms. Within hiring bins, low-investment firms earn higher returns than high-investment firms.

6.4 CAPM fails, Fama–French performs better

The CAPM cannot explain the hiring spread because market betas move in the wrong direction: high-hiring firms have higher market betas but lower returns. The Fama–French three-factor model does better, especially in value-weighted portfolios, suggesting that the hiring spread is related to priced risk not captured by the single market factor.

7 Model structure

7.1 Production technology

The firm uses capital and labor to produce output with a CES production function:

$$Y_t = Z_t X_t^{1-\nu} \left[a K_t^{1-1/\phi} + (1-a) N_t^{1-1/\phi} \right]^{\nu/(1-1/\phi)}.$$

- X_t is aggregate productivity.
- Z_t is firm-specific productivity.
- K_t is capital.

- N_t is labor.
- ϕ governs the elasticity of substitution between capital and labor.

7.2 Labor and capital accumulation

Labor evolves according to

$$N_{t+1} = (1 - \delta_n)N_t + H_t,$$

and capital evolves according to

$$K_{t+1} = (1 - \delta_k)K_t + I_t.$$

Hiring and investment are therefore state variables that determine the future production capacity of the firm.

7.3 Adjustment costs

The model includes both convex and nonconvex labor and capital adjustment costs.

- Labor adjustment costs include hiring, firing, training, screening, disruption, and separation costs.
- Capital adjustment costs include installation, planning, learning, production disruption, and costly reversibility.
- The adjustment cost function is asymmetric: hiring and firing can have different costs, and investment and disinvestment can have different costs.

The total adjustment cost is divided by a stochastic aggregate wedge S_t :

$$\Omega_t = \frac{C_{adj,t}^N + C_{adj,t}^K}{S_t}.$$

A positive S_t shock lowers the cost of adjusting labor and capital.

7.4 Aggregate shocks and stochastic discount factor

The model has two aggregate shocks:

- aggregate productivity shock Δx_{t+1} ;
- aggregate adjustment-cost shock Δs_{t+1} .

The stochastic discount factor is specified as

$$M_{t,t+1} = \exp(-r_f) \frac{\exp(-g_x \Delta x_{t+1} - g_s \Delta s_{t+1})}{E_t[\exp(-g_x \Delta x_{t+1} - g_s \Delta s_{t+1})]},$$

with $g_x > 0$ and $g_s < 0$.

7.5 Risk-premium equation

The model implies a two-factor risk-premium relation:

$$E[r_{t+1}^e] = \lambda_x \beta_x + \lambda_s \beta_s,$$

where $\lambda_x > 0$ is the price of aggregate productivity risk and $\lambda_s < 0$ is the price of adjustment-cost risk.

This equation is the heart of the paper’s economic mechanism. High-hiring firms have high positive exposure to adjustment-cost shocks. Since this shock has a negative price of risk, high-hiring firms earn lower expected returns.

8 Calibration and model fit

8.1 Targeted moments

The calibration targets aggregate asset-pricing moments, aggregate quantity moments, and firm-level hiring and investment moments.

- The model matches the average market excess return, Sharpe ratio, risk-free rate, and value premium.
- It matches important moments of hiring and investment rates, including volatility, skewness, persistence, and percentile ranges.
- The hiring and investment return spreads are not directly targeted; they are used to evaluate model performance.

8.2 Adjustment cost magnitudes

The implied adjustment costs are economically reasonable. The model implies that total realized labor adjustment costs are about 1.6 percent of annual sales and capital adjustment costs about 0.6 percent of annual sales. Marginal hiring costs are roughly 2.6 months of work, and marginal firing costs roughly 1.7 months of work.

8.3 Model performance

The baseline model reproduces the main empirical patterns.

- The model generates a low-minus-high hiring return spread of about 6.0 percent per year, close to the value-weighted data estimate of 5.6 percent.
- It generates investment and hiring return spreads in the two-way sorts.
- It reproduces the CAPM failure and the stronger Fama–French performance.

9 Economic mechanism

9.1 Why high-hiring firms have low expected returns

High-hiring firms are firms that recently received good idiosyncratic productivity shocks. Because productivity is persistent, these firms expect to keep expanding. Because expansion is costly, they are especially sensitive to the aggregate cost of adjusting labor and capital.

When the economy receives a positive adjustment-cost shock, adjustment becomes cheaper. High-hiring firms benefit disproportionately because they have more planned expansion. Their firm values rise strongly in these states. These states are high marginal-utility states in the model, so high-hiring firms provide a hedge and earn low expected returns.

9.2 Why productivity shocks do not explain the spread

High-hiring firms also have somewhat higher exposure to aggregate productivity shocks. Since productivity shocks have a positive price of risk, this force would make high-hiring firms riskier and raise their expected returns. This force goes in the wrong direction relative to the data. Therefore, the adjustment-cost shock is essential.

9.3 Why labor adjustment costs matter

If labor is freely adjustable, the model fails.

- Hiring rates become too volatile.
- The hiring return spread becomes too small or even negative.
- The cross-sectional return-predictability slopes weaken sharply.

The paper's conclusion therefore depends not merely on adding labor as an input, but on making labor costly to adjust.

10 Interpretation and caveats

10.1 Interpretation

The hiring return spread is interpreted as a risk premium, not as mispricing. High-hiring firms earn low future returns because they hedge adjustment-cost shocks, not because investors irrationally overvalue hiring firms.

10.2 Important caveats

- Employment data in Compustat are noisy and do not distinguish full-time, part-time, seasonal, or skill composition of labor.
- The stochastic discount factor is specified directly rather than derived from a fully specified household problem.
- The model uses a reduced-form adjustment-cost shock, which can stand in for multiple economic forces, such as investment-specific shocks, labor-market matching efficiency, or changes in the ease of hiring and investing.
- The empirical relation is predictive and equilibrium-based; it is not a causal estimate of hiring on returns.

11 Extracted figures and tables from the paper

11.1 Table 1: Hiring Portfolios and Table 2: Accounting Characteristics of the Hiring Portfolios

	EQUAL-WEIGHTED										VALUE-WEIGHTED									
	All Firms					All but Micro					All Firms					All but Micro				
	Low	2	5	9	High	Low	2	5	9	High	Low	2	5	9	High	Low	2	5	9	High
r	12.32	12.54	9.87	6.69	1.88	10.44	8.64	9.20	8.33	5.81	1.74	6.80	7.03	5.96	4.15	4.90	1.42	5.61		
β	2.96	3.51	5.03	1.76	.45	5.76	5.46	2.83	2.76	1.57	.41	3.48	2.47	2.20	1.75	1.52	.38	2.26		
SR	.48	.58	.50	.28	.07	.92	.40	.45	.44	.25	.06	.53	.39	.54	.36	.24	.06	.34		
Excess Returns																				
CAPM											Fama-French									
mean = 4.67											mean = 2.96									
α	6.57	7.06	4.72	.58	-4.75	11.32	2.80	3.80	3.14	-.42	-5.28	8.17	2.25	1.40	-.67	-.66	-4.78	7.03		
β	2.57	3.96	3.22	.31	-2.11	6.45	1.86	2.66	2.84	-.28	-2.69	4.28	1.69	1.13	-.07	-.48	-2.73	2.90		
δ	1.22	1.12	1.00	1.29	1.40	-.19	1.21	1.14	1.10	1.32	1.49	-.27	1.01	.96	.89	1.18	1.31	-.30		
γ	22.83	27.78	31.74	31.90	29.66	-5.32	30.45	29.36	37.97	43.88	37.30	-6.13	51.99	32.67	37.82	41.01	31.98	-4.96		
R^2	.57	.71	.77	.73	.69	.97	.78	.80	.85	.81	.77	.11	.78	.79	.82	.82	.77	.68		
mean = 2.50											mean = 1.15									
α'	2.27	3.03	1.61	-1.38	-6.32	8.59	-.53	.15	.68	-1.03	-5.06	4.53	.85	.14	.39	.71	-2.41	3.26		
β'	1.29	2.61	1.62	-1.12	-3.90	3.09	-.44	.15	.78	-.84	-3.35	3.00	.68	.11	.30	.58	-1.66	1.61		
δ'	1.02	1.03	1.01	1.09	1.17	-.14	1.17	1.16	1.08	1.17	1.27	-.11	1.05	1.04	.95	1.06	1.15	-.10		
γ'	1.16	.78	.65	.90	.98	.17	.56	.37	.36	.39	.69	-.13	.04	-.10	-.23	.22	.26	-.22		
R^2	.28	.38	.28	-.02	-.12	.40	.36	.48	.28	-.13	-.31	.68	.22	.26	.65	-.32	-.31	.73		

NOTE.—This table reports the average equal- and value-weighted excess stock returns and abnormal returns of 10 portfolios one-way sorted on hiring rate. We report portfolios 1 (low), 2, 5, 9, and 10 (high). The term r is the average annualized ($\times 1,000$) portfolio excess stock return. β is the beta coefficient and α is the intercept from monthly CAPM or Fama-French (1993) regressions, respectively, reported in annual percentages ($\times 1,000$). δ is the mean absolute pricing errors (average of absolute values of α or β across portfolios). γ is the mean absolute pricing errors (average of absolute values of α or β across portfolios). δ' and γ' are the portfolio market, SMB, and HML betas, respectively. L-H stands for the low-minus-high hiring portfolio. The table reports the equal-weighted returns across a sample that includes all the firms in the Compustat database from 1965 to 2010.

Table 1: Hiring Portfolios. The table shows that low-hiring firms earn substantially higher future returns than high-hiring firms. The CAPM cannot explain the spread, while the Fama-French model performs better.

TABLE 2
ACCOUNTING CHARACTERISTICS OF THE HIRING PORTFOLIOS

	A. DATA					B. MODEL				
	Low	2	5	9	High	Low	2	5	9	High
Hiring rate:										
HN_t	-.19	-.06	.03	.21	.44	-.63	-.31	-.20	-.05	.22
HN_{t+1}	-.01	-.01	.02	.07	.09	-.10	-.19	-.13	-.04	.12
Investment rate:										
IK_t	.14	.16	.19	.31	.37	-.23	-.05	-.01	.06	.28
IK_{t+1}	.15	.15	.19	.27	.29	-.14	-.04	.00	.07	.22
Productivity and profitability:										
TFP_t	.46	.50	.55	.59	.60	-.14	.02	.02	.03	.05
TFP_{t+1}	.48	.50	.54	.57	.58	-.10	.02	.02	.03	.05
ROA_t	-.01	.03	.06	.06	.05	-.06	.12	.16	.28	.53
ROA_{t+1}	.01	.04	.05	.05	.03	-.02	.13	.17	.28	.49
Valuation:										
KM_t	.49	.55	.41	.28	.25	.24	.83	.72	.57	.44
KM_{t+1}	.46	.52	.43	.31	.31	.15	.79	.68	.55	.47
$Size_t$	3.61	4.68	5.20	4.71	4.46	-.85	4.65	4.60	4.70	5.03
$Size_{t+1}$	3.82	4.86	5.41	4.79	4.52	-.70	4.56	4.54	4.68	5.08

NOTE.—This table reports the time-series averages of the following portfolio-level characteristics of 10 portfolios one-way sorted on hiring rate. We report portfolios 1 (low), 2, 5, 9, and 10 (high). HN is the hiring rate; IK is the investment rate; TFP is total factor productivity; ROA is return on assets (in the model, ROA is measured as profits scaled by the stock of physical capital); size is the log market capitalization; KM is the physical-capital-to-market-equity ratio. L-H stands for the low-minus-high hiring portfolio. The subscripts t and $t + 1$ stand for portfolio-level characteristics measured at the time of portfolio formation (t) or 1 year after portfolio formation ($t + 1$). The portfolio-level characteristics are computed as the median value of each characteristic across all firms in the portfolio in July of any given year. Panel A reports the statistics in the data from July 1965 to June 2010. Panel B reports the statistics using data simulated from the model, obtained as averages from 500 samples of simulated data, each with 3,600 firms and 600 monthly observations.

Table 2: Accounting Characteristics of the Hiring Portfolios. High-hiring portfolios are high-investment, high-productivity, high-profitability portfolios, consistent with an expansion interpretation.

11.2 Table 3: Hiring and Investment Portfolios and Table 4: Firm-Level Stock Return Predictability Regressions

TABLE 3
HIRING AND INVESTMENT PORTFOLIOS

HN	EQUAL-WEIGHTED										VALUE-WEIGHTED									
	All Firms IK					All low Micro IK					All Firms IK					All Firms IK				
	L	M	H	L-H	[t]	L	M	H	L-H	[t]	L	M	H	L-H	[t]	L	M	H	L-H	[t]
	Excess Returns r^e										Excess Returns r^e									
L	12.75	11.29	10.31	2.45	1.88	9.06	9.25	6.88	2.18	1.34	7.55	5.65	5.65	5.95	1.65					
M	11.97	9.78	8.34	3.63	2.08	9.88	8.15	6.81	3.04	1.64	5.88	6.27	5.75	-.57	-.14					
H	9.25	7.27	.87	8.35	4.01	8.51	6.25	1.10	7.40	2.97	7.46	4.44	-1.09	8.55	2.90					
L-H	3.52	3.98	9.44			.55	2.99	5.77			.10	1.20	4.72							
[t]	2.74	3.49	5.50			.51	2.59	2.77			.06	.78	1.83							
	CAPM α										CAPM α									
	m.a.e. = 4.79					m.a.e. = 5.00					m.a.e. = 2.51					m.a.e. = 2.51				
L	7.97	5.79	4.40	2.97	2.30	3.62	5.94	.86	2.74	1.75	3.11	1.28	-1.21	4.82	2.35					
M	7.12	4.75	2.28	4.84	2.98	4.90	5.16	.69	4.21	2.41	1.55	1.94	.21	1.54	.57					
H	5.81	1.26	-6.03	9.84	4.99	3.09	.56	-4.19	9.27	3.96	2.68	-.34	-8.00	10.68	3.81					
L-H	3.56	4.23	10.43			.55	5.38	7.67			.13	1.62	6.29							
[t]	2.84	5.73	6.24			.48	2.95	3.51			.28	1.06	2.50							
	Fama-French α^f										Fama-French α^f									
	m.a.e. = 2.14					m.a.e. = 1.25					m.a.e. = 1.63					m.a.e. = 1.63				
L	2.59	1.95	1.62	.97	.85	-.71	.76	-.65	-.88	-.06	.95	.69	-1.17	2.12	1.06					
M	2.70	1.89	.88	1.81	1.46	1.58	.88	.59	.99	.75	.21	2.50	5.00	-2.79	-1.37					
H	-.04	-1.29	-6.56	6.32	5.96	.26	-1.10	-4.74	5.00	2.84	1.59	-.35	-4.17	5.76	2.56					
L-H	2.63	3.16	7.97			-.97	1.86	4.11			-.65	.34	5.00							
[t]	2.06	2.77	2.25			-.97	1.76	2.42			-.44	.25	1.43							

NOTE.—This table reports the average equal and value-weighted excess stock returns and abnormal returns of nine portfolio two-way sorted on hiring rate (HN) and investment rate (IK). The sorting on the hiring rate is reported across rows (L, M, and H) and the sorting on the physical capital investment rate is reported across columns (L, M, and H). L-H stands for the low-investment-high hiring portfolio (across rows) or the low-investment-high investment portfolio (across columns). r^e is the average annualized ($\times 1,200$) portfolio excess stock return; [t] are heteroscedasticity and autocorrelation consistent t -statistics (Newey-West); α and α^f are the portfolio average abnormal returns, obtained as the

Table 3: Hiring and Investment Portfolios. The two-way sorts show that hiring predicts returns within investment portfolios and investment predicts returns within hiring portfolios.

11.3 Table 5: Parameter Values of the Baseline Model and Table 6: Target Moments

TABLE 5
PARAMETER VALUES OF THE BASELINE MODEL

Parameter	Symbol	Value
Technology: general:		
Weight of physical capital in the production function	α	.36
Returns to scale	θ	.85
Elasticity of substitution between capital and labor	ϕ	.50
Rate of depreciation for capital	δ_k	.01
Quit rate of labor	δ_n	.01
Fixed operating cost	f	.01
Technology: adjustment costs:		
Convex parameters in capital adjustment cost	c_k^+ / c_k^-	3.1/34
Convex parameters in labor adjustment cost	c_n^+ / c_n^-	1.2/1.
Nonconvex parameters in capital adjustment cost	b_k^+ / b_k^-	.04/.0
Nonconvex parameters in labor adjustment cost	b_n^+ / b_n^-	.16/.2
Stochastic processes:		
Multiplicative coefficient on wage rate process	τ_1	.00
Sensitivity of the wage rate to aggregate productivity	τ_2	.9
Average growth rate of aggregate productivity	μ_x	.013/1
Conditional volatility of aggregate productivity	σ_x	.05
Average level of firm-specific productivity	$\bar{z}$	-3.4

Table 5: Parameter Values of the Baseline Model. The calibration specifies technology, adjustment-cost, stochastic-process, wage, and discount-factor parameters.

LABOR HIRING, INVESTMENT

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TABLE 4
FIRM-LEVEL STOCK RETURN PREDICTABILITY REGRESSIONS

	CROSS-SECTIONAL REGRESSIONS ($N = 1,569$)				POOLED OLS REGRESSIONS ($N = 65,805$)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HN_{t-1}	-.89	-.75	-.71	-.48	-.18	-.15	-.13	-.07
[t]	-5.93	-5.30	-3.38	-3.09	-5.87	-5.67	-3.36	-2.31
IK_{t-1}		-.52		-.54		-.16		-.23
[t]		-2.40		-1.90		-3.39		-3.63
$\text{Micro} \times HN_{t-1}$			-.11	-.24			-.03	-.07
[t]			-.49	-1.20			-.88	-2.14
$\text{Micro} \times IK_{t-1}$				.02				.13
[t]				.10				2.38

NOTE.—This table reports the estimation results from several variations of stock return predictability regressions of the form

$$r_i^e = a + b \times HN_{t-1} + c \times IK_{t-1} + d \times \text{Micro} + e \times \text{Micro} \times HN_{t-1} + f \times \text{Micro} \times IK_{t-1} + e_u,$$

in which r_i^e is the firm i stock return, HN_{t-1} and IK_{t-1} are the lagged values of firm i 's hiring and investment rates, and Micro is a dummy variable that is equal to one if firm i is a micro cap firm at time $t - 1$. Micro caps are defined as the firms with size (market value) that is below the bottom 20th percentile of the cross-sectional size distribution of NYSE firms in June of each year. Two alternative methodologies are used to estimate the regression. Columns 1–4 report the estimated average slope in the previous equation from Fama-MacBeth (1973) cross-sectional regressions estimated at the monthly frequency; [t] are heteroscedasticity and autocorrelation consistent t -statistics (Newey-West); N is the average number of firms in each cross section. Columns 5–8 report the estimated slope coefficients in the previous equation obtained by pooled OLS regressions in which r_i^e is firm i 's compounded annual stock return from July of year t to June of year $t + 1$. The regression includes both year and firm fixed effects; [t] are t -statistics computed from standard errors clustered by firm and year; and N is the number of firm-year observations

Table 4: Firm-Level Stock Return Predictability Regressions. Hiring and investment have negative predictive slopes for future returns, even when both variables enter jointly.

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TABLE 6
TARGET MOMENTS

Moment	Data	Model
Asset prices:		
Average stock market excess return (%)	4.85	4.84
Sharpe ratio of aggregate stock market	.31	.31
Average real risk-free rate (%)	1.65	1.65
Value premium (%)	6.73	5.46
Average aggregate capital-to-market-equity ratio	.62	.57
Real quantities and input prices:		
Standard deviation of aggregate profits	.14	.14
Standard deviation of wage rate	1.40	1.32
Average firm-level wage-bill-to-sales ratio	.46	.42
Correlation of aggregate payout and output	.38	.33
Firm-level hiring and investment rate: cross section:		
Standard deviation of HN (dispersion)	.23	.22
Standard deviation of IK (dispersion)	.21	.17
Correlation (HN_t, HN_{t-2})	.10	.10
Correlation (IK_t, IK_{t-2})	.37	.23
Skewness of HN	.74	.71
Skewness of IK	1.77	1.60
95th minus 5th percentile range of HN	.66	.73
95th minus 5th percentile range of IK	.61	.54
Firm-level hiring and investment rate: time series:		
Standard deviation of HN	.26	.24
Standard deviation of IK	.23	.22
Correlation (HN_t, HN_{t-2})	.10	.13
Correlation (IK_t, IK_{t-2})	.38	.20
Skewness of HN	.74	.71
Skewness of IK	1.77	1.60
95th minus 5th percentile range of HN	.66	.73
95th minus 5th percentile range of IK	.61	.54

Table 6: Target Moments. The model is disciplined by asset-pricing moments, aggregate quantities, and firm-level hiring and investment moments.

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MEASURE OF ADJUSTMENT COSTS	LABOR			CAPITAL		
	$H > 0$	$H < 0$	All	$I > 0$	$I < 0$	All
1. Average realized total (% annual sales)			1.63			.55
2. Nonconvex (% annual sales)	1.32	1.69		.32	.70	
3. Marginal convex (% annual wage)	21.47	14.16				
4. Marginal convex (months of work)	2.58	1.70				
5. Marginal convex (per \$100 of investment)				\$6.58	\$8.70	

Table 7: Implied Adjustment Costs. The model-implied labor and capital adjustment costs are economically plausible relative to existing empirical estimates.

TABLE 8
HIRING AND INVESTMENT PORTFOLIOS IN SIMULATED DATA

A. ONE-WAY SORTED										B. TWO-WAY SORTED				
	Low	2	5	9	High	L-H	$\overline{HN}$	L	M	H	$\overline{LH}$	$\langle f \rangle$		
	Excess Returns													
σ^2	8.08	7.16	5.60	5.22	2.06	6.01	L	8.04	6.01	4.22	3.82	6.11		
σ	4.02	3.33	2.41	2.28	1.01	8.60	M	7.02	5.25	2.28	3.03	6.11		
SR	.57	.48	.35	.30	.13	1.61	L-H	1.56	1.41	2.11	3.43	7.21		
								2.56	1.41	2.11	3.85			
								$\langle f \rangle$						
	CAPM: $\beta_{\text{CAPM}} = 1.66$													
σ^2	3.79	2.55	.65	-1.74	-2.90	6.60	L	3.73	1.30	-1.56	4.27	7.38		
σ	8.74	6.08	1.66	-4.55	-7.59	9.94	M	2.51	2.28	-1.64	4.15	6.86		
SR	.88	.65	1.62	1.63	1.03	1.14	L-H	-2.12	-1.55	-1.55	3.97	7.10		
$\overline{R}$	91.50	100.93	133.51	130.08	118.85	-91.2	L-H	2.12	2.84	2.29	2.29			
	.95	.96			.57	.10	$\langle f \rangle$	4.02	4.29	4.29	4.24			
	Fama-French: $\beta_{\text{FF}} = .58$													
σ^2	.63	.11	-.15	-.03	-.04	.07	L	.57	.01	-.06	.83	1.05		
σ	1.55	.26	-.34	-.07	-.10	1.23	M	.05	-.20	-.03	.97	.11		
SR	.96	1.01	1.02	.99	.97	-.01	L-H	.05	-.10	-.06	.90	.01		
$\overline{R}$	.29	.21	.15	.17	.30	.58	L-H	.05	.11	.00				
β	.54	.43	.10	-.29	-.47	1.01	$\langle f \rangle$	1.05	.17	.01				
β	.97	.97	.98	.98	.98									

Table 8: Hiring and Investment Portfolios in Simulated Data. The simulated model reproduces the hiring spread, the investment spread, CAPM pricing errors, and improved Fama–French fit.

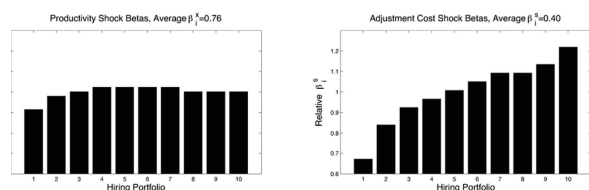


Figure 1: Risk Exposures of the 10 Hiring Portfolios. Exposure to productivity shocks is nearly flat across hiring portfolios, while exposure to adjustment-cost shocks rises strongly with hiring.

TABLE 9
SELECTED DATA VERSUS MODEL-IMPLIED MOMENTS ACROSS ALTERNATIVE CALIBRATIONS

	SELECTED DATA VERSUS MODEL-DERIVED MOMENTS ACROSS ALTERNATIVE CALIBRATIONS													
	QUANTITIES					ASSET PRICES								
	SD	Skewness	Market	Value	HN Spread			IK Spread			XS Slopes			
					One-Way	Two-Way	Two-Way	One-Way	Two-Way					
SPECIFICATION	HN	IK	HN	IK	f	α	f	α	f	α	HN	IK		
0. Data	24	23	74	177	4.85	6.73	5.64	7.03	2.01	2.78	4.04	5.61	-48	-54
1. Baseline model	24	22	71	160	4.84	5.46	6.01	6.69	1.67	2.11	4.03	4.27	-50	-50
2. Low price of risk of adjustment cut-shock ($\gamma_c = 7$)	24	21	69	156	5.70	8.1	5.8	2.15	3.88	1.16	12	85	-12	20
3. Low price of risk of aggregate productivity shock ($\gamma = 1.5$)	24	21	59	156	3.38	-2.97	5.08	4.86	1.29	1.21	3.53	3.54	-26	-49
4. No labor concern adjustment ($\zeta_c = \zeta_n = c_f = 0.015$)	42	25	14	181	5.08	5.21	4.94	5.86	1.78	2.37	3.16	3.40	-17	-49
5. No labor concern adjustment ($\zeta_c = \zeta_n = c_f = 0.031$)	25	21	64	155	4.99	10.7	1.00	2.87	1.01	1.61	-0.02	68	-62	62
6. Frictionless labor ($\zeta_c = \zeta_n = \zeta_f = 0.4$, $\theta = 0.01$, $f = 0.04$)	40	24	19	178	4.83	-13.5	-8.2	1.24	1.13	1.80	-15.7	-7	-7	-61
7. No fixed operating costs ($\theta = 0$)	25	22	68	159	11.00	-3.45	4.18	1.34	1.64	2.22	6.05	5.61	-29	-95

^aNote.—This table presents selected moments of the firm-level labor hiring rate (H_{it}), the physical capital investment rate (K_t), and asset prices both in the real data (specification O) and simulated by the simulation of the model under alternative specifications (specifications A–G). The table reports the following moments: the time-series standard deviation (SD) of firm-level H_{it} and K_t ; the time-series average of the cross-sectional skewness of H_{it} and K_t ; the average excess return on the value-weighted aggregate stock market (market); the average return of the high-minus-low book-to-market decile portfolio (value); H_{it} spread is the labor hiring return spread across 10 portfolios one-way sorted on hiring (one-way) or the average hiring spread across investment portfolios in nine portfolios two-way double-sorted on hiring and investment portfolios (two-way); K_{it} spread is the average investment return spread across firms; H_{it} slope is the labor hiring return spread across five portfolios double-sorted on hiring and investment portfolios (two-way); X slopes are the H_{it} and K_{it} slopes on the corresponding market returns. All variables are expressed as percentages unless otherwise stated. The sample period is from 1980:Q1 to 2007:Q4. Data source: Compustat, CRSP, and Fama-MBl.

Table 9: Selected Data versus Model-Implied Moments across Alternative Calibrations. The hiring spread depends critically on adjustment-cost risk and labor adjustment costs.

11.6 Figure 2: Impulse Responses to an Adjustment-Cost Shock and Figure 3: Impulse Responses to a Productivity Shock

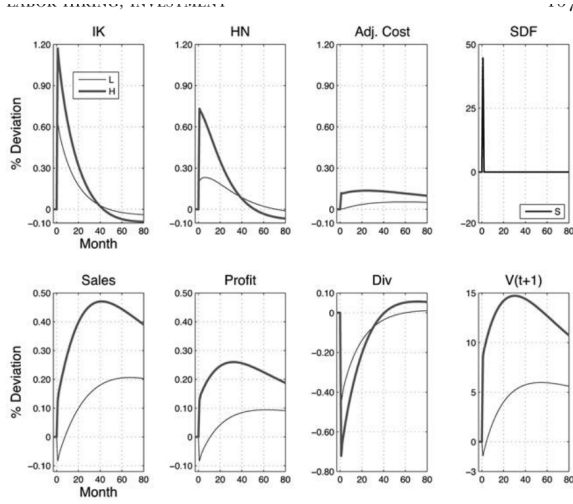


FIG. 2.—Impulse responses of selected endogenous variables in the baseline calibration of the model to a positive aggregate adjustment cost shock (reduction in marginal adjustment cost). The responses are measured in percentage point deviations relative to the long-run average values (time detrended, when applicable). To generate the response of a high-productivity (H) firm, we add a +3 percent firm-specific productivity shock. To generate the response of a low-productivity firm (L), we add a -3 percent firm-specific

Figure 2: Impulse Responses to a Positive Aggregate Adjustment-Cost Shock. High-productivity/high-hiring firms benefit more from lower adjustment costs, so their continuation values rise more.

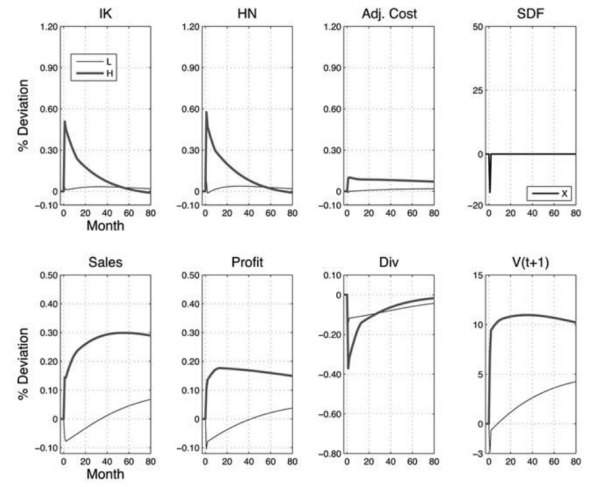


FIG. 3.—Impulse responses of selected endogenous variables in the baseline calibration of the model to a 3 percent positive aggregate productivity shock. The responses are measured in percentage point deviations relative to the long-run average values (time detrended, when applicable). To generate the response of a high-productivity (H) firm, we add a +3 percent firm-specific productivity shock. To generate the response of a low-productivity firm (L), we add a -3 percent firm-specific productivity shock. The frequency

Figure 3: Impulse Responses to a Positive Aggregate Productivity Shock. Productivity shocks generate a response that would make high-hiring firms riskier, so they cannot explain the low expected returns of high-hiring firms.

12 How to read the paper quickly

12.1 For the empirical result

Read Section II and Tables 1–4. The key pattern is that hiring predicts future returns even after controlling for investment.

12.2 For the model

Read Section III and the risk-premium equation. The key conceptual device is the aggregate adjustment-cost shock with a negative price of risk.

12.3 For the mechanism

Read Section VI and Figures 1–3. The core message is that high-hiring firms have larger exposure to adjustment-cost shocks, and this exposure makes them hedges.

13 Conclusion

The paper’s main contribution is to make labor hiring an asset-pricing state variable. Hiring is not only a labor-demand outcome; it is also a forward-looking firm investment decision that

reveals exposure to macroeconomic adjustment-cost risk. Empirically, high-hiring firms earn lower future returns. Theoretically, this occurs because high-hiring firms are expanding firms that benefit disproportionately when adjustment costs fall. The result broadens investment-based asset pricing from capital adjustment costs to labor-market frictions and shows that financial-market data can help quantify the economic importance of labor adjustment frictions.